

Jillian Brown

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Education

Master of Science in Natural Language Processing

Sep 2025 - June 2027

University of Lorraine | University of Groningen

- Erasmus Mundus Language & Communication Technology Scholar
- Double Degree Program

Bachelor of Arts in Global Studies & Linguistics

Aug 2020 - Dec 2024

South Carolina Honors College, University of South Carolina, *Columbia, SC, USA*

- *Thesis*: "Pragmatic & Discourse Particles of Singaporean English"
- *GPA*: 3.98

Professional Experience

Research Assistant, LORIA Laboratory | University of Lorraine

January - June 2026

- Designed three-stage experimental validation pipeline for evaluation of acoustic indicators of depression in speech through statistically and ML-based methods.
- Built a reproducible Python audio preprocessing pipeline integrating multiple third-party toolboxes (librosa, openSMILE, Praat), showcasing practical audio engineering skills for NLP application.
- Applied linear mixed-effects models in R to isolate confounding variables in speech feature data.
- Developed and critically interpreted a binary classifier for depression detection using cross-validated logistic regression, with attention to class imbalance, clinical constraints, and model limitations.

Software Developer, ScholasTech | Wordification

May 2024 - Present

- Engineered automated tools to transform external linguistic data into structured inputs, increasing data processing efficiency for core software systems.
- Developed scalable full-stack features using Redwood.js and React.js, ensuring high-performance integration between frontend UI and backend services.
- Optimized relational database schemas and SQL queries, reducing data retrieval latency and improving the accuracy of real-time reporting.
- Cross-collaborated with educators and engineers to bridge the gap between academic theory and technical implementation, delivering user-centric pedagogical tools.

Project Experience

DetectErreur

Sep 2025 - Jan 2026

Educational Technology Tool developed at University of Lorraine, *Nancy, France*

- Designed text processing algorithm using Python, spaCy, and Mistral to detect and correct errors in French learner writing.
- Employed prompt engineering and retrieval augmented generation (RAG) of small language models to generate educational feedback to student users.

Essential Skills

- **Programming & Web Development**: Python, R, JavaScript, CSS/HTML
- **Tools & Frameworks**: SQL, React, Git, PyTorch, Redwood.js, Audacity, Praat
- **Skills**: Statistical and ML Model Evaluations, Model Fine-Tuning, App Development, Retrieval Augmented Generation, Web Scraping, Prompt Engineering, Large Data Sets